



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

B.Sc. PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	SYNTHESIS OF NANOMATERIALS				
Type of Course	Vocational Minor (GROUP I: TECHNIQUES IN MATERIALS PHYSICS)				
Semester	II				
Academic Level	100 - 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	PHY1VN101- Introductory Materials Science				
Course Summary	This course gives an introduction to the fascinating world of nanomaterials and diverse synthesis methods.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Define and classify nanomaterials and explain size-dependent properties.	U	F	Instructor-created exams / Quiz

CO2	Analyze various physical and chemical methods for nanomaterial synthesis, including their advantages, limitations, and applications.	An	C	Instructor-created exams / Home Assignments
CO3	Explain nanofabrication techniques: Grasp the concepts and applications of different nanolithography techniques like electron beam and photonic methods.	Ap	C	Seminar Presentation / Group Tutorial Work
CO4	Select appropriate synthesis methods: Analyze material requirements and choose suitable synthesis methods for specific applications.	Ap	P	Instructor-created exams / Home Assignments
CO5	Perform basic nanomaterial synthesis: Conduct laboratory experiments to prepare nanomaterials using different techniques learned.	E	P	Practical Assignment/ Observation of Practical Skills
CO6	Work collaboratively: Successfully participate in team-based projects and experiments related to nanomaterials.	E	P	Practical Assignment / Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	Introduction to Nanomaterials		10	15
	1	Definition and classification of nanomaterials (0D, 1D, 2D, 3D),	2	
	2	Specific surface area of nanomaterials	2	

	3	Size-dependent properties of nanomaterials (Thermal, Electrical, Mechanical, Magnetic, Optical)	6	
	Sections 1.1 to 1.4, 2.1 to 2.3, 3.1 to 3.7, Book 1			
II	Synthesis of Nanomaterials (Physical Methods)		10	15
	4	High energy ball milling, melt mixing,	2	
	5	Physical vapour deposition	1	
	6	Ionized cluster beam deposition,	1	
	7	laser ablation, laser pyrolysis		
	8	Sputter deposition, chemical vapour deposition, electric Arc deposition	3	
	9	Ion beam technique, molecular beam epitaxy.	3	
	Section 3.1 to 3.8, Book 2			
III	Synthesis of Nanomaterials (Chemical Methods)		10	25
	10	Synthesis of metal nanoparticles by colloidal route	2	
	11	Synthesis of semiconductor nanoparticles by colloidal route	3	
	12	Sol-gel method,	1	
	13	Hydrothermal synthesis	2	
	14	Sonochemical synthesis	1	
	15	Microwave synthesis	1	
	Section 4.1 to 4.5, 4.8 to 4.11, Book 2			
IV	Synthesis of Nanomaterials (Other Methods)		15	15
	16	Synthesis Using Microorganisms	2	
	17	Synthesis Using Plant Extracts	2	
	18	Synthesis of Nanoparticles Using DNA	2	

	19	Nanolithography, Lithography Using Photons, Use of X-rays in Lithography	5	
	20	Lithography Using Particle Beams, Electron Beam Lithography	2	
	21	Ion Beam Lithography	1	
	22	Neutral Beam Lithography	1	
	Section 5.1 to 5.5, 9.1 to 9.3.3, Book 2			
V	PRACTICALS		30	
	1	Preparation of samples using two different techniques discussed in Module II - IV		
Books and References:				
1) Nanomaterials and Nanocomposites: Synthesis, Properties, Characterization Techniques, and Applications by Rajendra Kumar Goyal (Book 1)				
2) Nanotechnology_ Principles and Practices by Sulabha K. Kulkarni, 3rd Edition (Book 2)				
3) Springer Handbook of Nanomaterials by Robert Vajtai				

Mapping of COs with PSOs and POs :

	PS O1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	2	0	1	0	0	0	3	3	2	0	3	3	1
CO 2	1	2	1	2	3	2	3	3	2	2	3	3	2
CO 3	1	1	3	2	2	2	3	3	2	2	3	3	2
CO 4	2	1	2	3	3	2	3	3	2	3	3	3	2
CO 5	3	2	2	3	3	3	3	3	3	3	3	3	3
CO 6	3	3	2	2	2	2	3	3	2	3	3	3	3

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low

2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓			✓
CO 3	✓			✓
CO 4		✓		✓
CO 5		✓		✓
CO 6			✓	